Project Name: Indoor Human Detection and Tracking under Temporary Occlusion

Nowadays, Human Tracking technology is widely used in the fields of healthcare, smart home and industrial warehouses. However, the precision and accuracy of the tracking are significantly influenced when the target is temporarily lost due to obstacles, occlusion, or changes in the person's movements. This project will study indoor human detection and tracking under temporary occlusion using radar measurements and Kalman filtering. A simplified indoor environment will be modeled with a single moving person, a radar sensor, and random obstacles that can partially or fully block the line of sight between the target and the sensor. The target motion and radar measurements will be simulated with measurement noise and intermittent loss of observations during occlusion.

This problem naturally involves both detection and estimation. A classical detection method will be used to determine whether the target is present at certain confidence at each time step. When a valid measurement is available, it will be used to update a Kalman-based state estimator. During temporary target loss, the estimator will try to predict the target state based on the motion model until the target can be detected again.

The main objective is to investigate how intermittent observations will affect target detection probability, estimation uncertainty, and tracking accuracy. Different occlusion durations, noise levels, and detection conditions will be evaluated using metrics such as probability of detection, false alarm probability, and tracking RMSE.